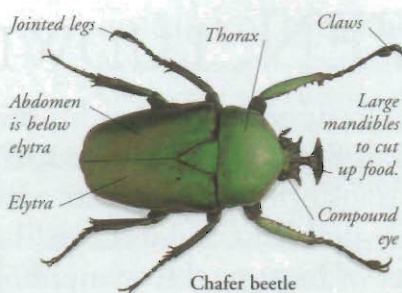


BEETLES



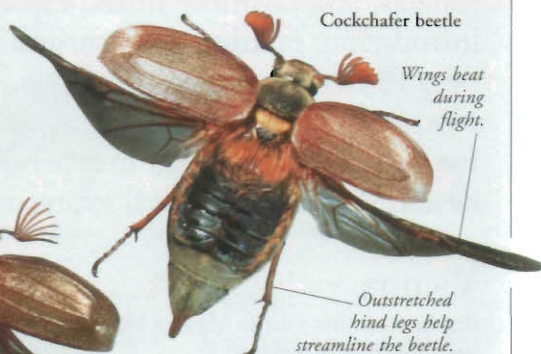
THERE ARE AT LEAST 350,000 types of beetle. They make up 30 per cent of all animals and 40 per cent of all insects. They range in size from the 2 mm (0.08 in) long battle d'or beetle to the giant timber beetles, which grow up to 150 mm (6 in) long. Beetles live almost everywhere, from hot deserts to snowy mountain tops, but they are most numerous in the tropics. They eat a wide range of food, including crops, and are considered pests, but they perform a valuable role by breaking down dead animals and plants and returning the nutrients to the soil.



Features of a beetle

Beetles have three body parts – the head, thorax, and abdomen. They have compound eyes, and antennae used for touch and smell. Their forewings have developed into hard wing cases, or elytra, which protect the hind wings. The wings, elytra, and six legs are fixed to the thorax.

2 The beetle opens its wing cases. These act as stabilizers, similar to the tail wings of an aeroplane. The delicate hind wings unfold and provide the main force in flight.



How a beetle flies

1 Large beetles, such as this cockchafer beetle, take a few seconds to get airborne. First the beetle pumps air into its body by expanding its abdomen.



3 The cockchafer beetle pushes off with its legs and starts to beat its hind wings. Within a few seconds the hind wings reach the 200 beats per second needed for take-off, and the wing cases help to provide lift. During flight, the beetle uses 100 times more oxygen than it does at rest.

Wood-boring beetles

Some beetles remain as larvae for many years. Jewel beetle larvae may live in wood for over 40 years. They eat the wood, making tunnels through it, leaving small holes.



Jewel beetle

Reproduction

Most beetles undergo complete metamorphosis. Larvae hatch from eggs laid by an adult female and are the main feeding stage in a beetle's development. Once the larvae have finished growing, they turn into pupae. Inside, they change, or metamorphose, into the adult beetle that will eventually emerge from the pupa.



Fighting

Male beetles often fight with each other over a possible mate. They use their mandibles (mouthparts) as weapons. Stag beetles have huge, but not very powerful mandibles, which they use mainly to impress rivals. Despite their size, the mandibles do little harm. In this way, fighting is more symbolic, and both beetles live to fight and mate again.

The beetle clasps his rival in his huge jaws and tries to throw it on its back.



Feeding

Beetles' feeding habits, like beetles themselves, are diverse. Many, including spider beetles, feed on decaying leaf litter; others consume both living and dead wood. Some beetles, such as tiger beetles, actively hunt for live food. Scavengers, such as hide beetles, feed on rotting vegetation, dead animals, and dung. Some beetles, for example, rove beetles, are even parasites, living on creatures such as bats.

Ladybirds

Ladybirds are found worldwide. They prey on small insects, such as aphids and scale bugs. In this way they are helpful animals to have in the garden and can be used to control pests instead of using polluting chemicals.



Defence

Well-armoured external skeletons and camouflage protect many beetles from predators. The bombardier beetle has an ingenious method of defence. It ejects a hot mixture of potent chemicals from its rear with an audible pop.



Bombardier beetle

Water beetles

Many beetles live in water. Diving beetles use their oar-like legs to push themselves through the water after their prey. Whirligig beetles scavenge for food that floats on the water's surface. They have special eyes that are split in two. One half looks downwards for fish, while the top half scans the air for predatory birds.



Whirligig beetle

COCKCHAFER BEETLE

SCIENTIFIC NAME

Melolontha melolontha

ORDER

Coleoptera

FAMILY

Scarabaeidae

DISTRIBUTION

Europe and western Asia

HABITAT

Gardens and woods

DIET

Adult feeds on sap and nectar; the larvae feed on the roots of plants, such as rose and oak

SIZE

Larvae: 4 cm (1.6 in) in length; adults: 2–3 cm (0.8–1.2 in) long

LIFESPAN

Larvae take about 2 years to become adults; adults live for about 2–3 months

FIND OUT MORE

ARTHROPODS

DESERT WILDLIFE

GRASSLAND WILDLIFE

INSECTS

WOODLAND WILDLIFE